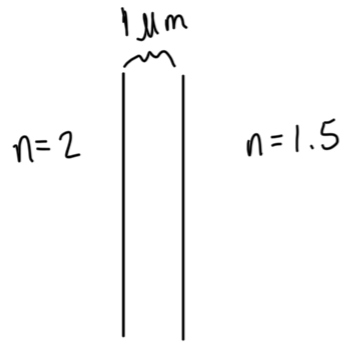
4.2



200 nm - 800 nm

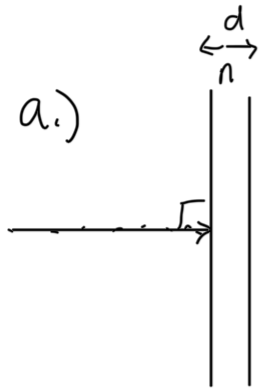
$r_s^{tot} =$

$$r_s = \begin{bmatrix} r_s^{0\to1} & r_s^{0\leftarrow1} \\ r_s^{1\to2} & r_s^{1\leftarrow2} \end{bmatrix}$$

$$t_s = \begin{bmatrix} t_s^{0\to1} \\ t_s^{1\to2} \end{bmatrix}$$

4.4

a.)



$T^{tot} + R^{tot} = 1$

$$r^{1\to2} = r^{0\leftarrow1} = -r^{0\to1} = \frac{n-1}{n+1}$$

$$R = |r|^2 = \left(\frac{n-1}{n+1}\right)^2 \qquad T = 1 - \left(\frac{n-1}{n+1}\right)^2$$

$$T^{tot} = \frac{T_{01}T_{12}}{\left(1 - r^{0\leftarrow1} r^{1\to2} e^{i2kdn}\right)^2} \qquad k = \frac{2\pi}{\lambda}$$

$$R = r^2 = r^{0\leftarrow1} r^{1\to2}$$

$$T_{01} = T_{12} = 1 - R$$

$$\hookrightarrow T^{tot} = \frac{(1-R)^2}{\left(1 - R\cos(2kd) - iR\sin(2kd)\right)^2} \qquad \frac{(1-R)^2}{|1 - Re^{i2kdn}|^2}$$

$$1 - R\cos(2kd) - i\cancel{R\sin(2kd)} - R\cos(2kd) + R^2\cos^2(2kd) + i\cancel{R^2\cos(2kd)\sin(2kd)}$$
$$+ i\cancel{R\sin(2kd)} - \cancel{iR^2\cos(2kd)}\sin(2kd) - i^2R^2\sin^2(2kd)$$

$$\hookrightarrow 1 - 2R\cos(2kd) + R^2\cos^2(2kd) + R^2\sin^2(2kd)$$

$$\Rightarrow \quad 1 - 2R\cos(2kd) + R^2 \qquad \rightarrow R^2$$

$$\cos(2kd) = 1 - 2\sin^2(kd)$$

$$= 1 - 2R + 4R\sin^2(kd) + R^2 \Rightarrow R^2 - 2R + 1 + 4R\sin^2(kd)$$

$$\Rightarrow (1-R)^2 + 4R\sin^2(kd)$$

$$T_{tot} = \frac{(1-R)^2}{(1-R)^2 + 4R\sin^2(kdn)} = \frac{1}{1 + \frac{4R}{(1-R)^2}\sin^2(kdn)}$$

$$R = \left(\frac{n-1}{n+1}\right)^2 \qquad 1 - R = 1 - \frac{(n-1)^2}{(n+1)^2} = \frac{(n^2+2n+1) - (n^2-2n+1)}{(n+1)^2} = \frac{4n}{(n+1)^2}$$

$$\frac{\cancel{4}(n-1)^2}{\cancel{(n+1)^2}} \cdot \frac{(n+1)^{\cancel{4}\,2}}{\cancel{16}_{4}\, n^2} = \frac{((n-1)(n+1))^2}{4n^2} = \frac{(n^2-1)^2}{4n^2}$$

$$T_{tot} = \frac{1}{1 + \frac{(n^2-1)^2}{4n^2}\sin^2\left(\frac{2\pi n d}{\lambda}\right)} \qquad kdn = \frac{2\pi}{\lambda} dn$$

b.) $n = 1.5$

$$T_{tot} = \frac{1}{1 + \frac{25}{144}\underbrace{\sin^2\left(3\pi \frac{d}{\lambda}\right)}_{1}} \Rightarrow T_{max} = \frac{1}{\frac{169}{144}} \approx 85.2\%$$

c.) $d = 150\ \mu m$

$$\sin^2\left(3\pi \frac{150e^{-6}}{\lambda}\right) = 0 \quad \Rightarrow 450\pi e^{-6}$$

$$\frac{\quad}{\lambda} = m\pi$$

$$\Rightarrow \lambda = \frac{450e-6}{m}$$

P4.8

$R = 0.9$ $\quad d = 0.5e-2$ m

$T = 0.05$ $\quad n_1 = 1$

$A = 0.05$ $\quad \lambda = 587e-9$ m

$$T_{max} = \frac{T^2}{(1-R)^2}$$

$$T_{min} = \frac{T^2}{(1+R)^2}$$

4.17

a.) $r^{0\to 1} = r^{0\leftarrow 1} \Rightarrow \begin{bmatrix} E_{0\to} \\ E_{0\leftarrow} \end{bmatrix} = \begin{bmatrix} 0 & -i/n_1 \\ -in_1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ n_g \end{bmatrix}$

$= \begin{bmatrix} -n_g i/n_1 \\ -i n_1 \end{bmatrix}$ $\quad Y = \frac{E_{0\leftarrow}}{E_{0\to}} = \frac{n_1}{n_g/n_1} = \frac{n_1^2}{n_g}$ $\quad r_s = \frac{1-Y}{1+Y} = \frac{1-\frac{n_1^2}{n_g}}{1+\frac{n_1^2}{n_g}} = \frac{n_g - n_1^2}{n_g + n_1^2}$

$R = |r_s|^2 \Rightarrow R = \left(\frac{n_g - n_1^2}{n_g + n_1^2}\right)^2$

b.)

$$M = \begin{bmatrix} 0 & \frac{1}{n_1} \\ n_1 & 0 \end{bmatrix} \begin{bmatrix} 0 & \frac{1}{n_2} \\ n_2 & 0 \end{bmatrix} = \begin{bmatrix} \frac{n_2}{n_1} & 0 \\ 0 & n_1/n_2 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{n_2}{n_1} & 0 \\ 0 & \frac{n_1}{n_2} \end{bmatrix} \begin{bmatrix} 1 \\ n_g \end{bmatrix} = \begin{bmatrix} n_2/n_1 \\ n_g \frac{n_1}{n_2} \end{bmatrix}$$

$$Y = \frac{\frac{n_g n_1}{n_2}}{\frac{n_2}{n_1}} = \frac{n_g n_1^2}{n_2^2} \qquad r_s = \frac{1-Y}{1+Y} = \frac{1 - \frac{n_g n_1^2}{n_2^2}}{1 + \frac{n_g n_1^2}{n_2^2}}$$

$$\Rightarrow r_s = \frac{n_2^2 - n_g n_1^2}{n_2^2 + n_g n_1^2} \Rightarrow R = \left(\frac{n_2^2 - n_g n_1^2}{n_2^2 + n_g n_1^2}\right)^2$$

c.) ZnS: $n = 2.32$

CeF: $n = 1.63$

MgF: $n = 1.38$

$$R = \left(\frac{n_2^2 - 1.5 n_1^2}{n_2^2 + 1.5 n_1^2} \right)^2 \qquad R = \left(\frac{1.5 - n_1^2}{1.5 + n_1^2} \right)^2$$

make denom as small as possible. $\Rightarrow n_1 = 1.38$

since $n_2^2 < 1.5 n_1^2$ $\Rightarrow n_2 = 1.63$ using scheme b.

n_1 smallest

n_2 next smallest